Cooling experiment report

Comparison of insulated and uninsulated vessels

We compare how two vessels cool from 80 degrees C in a room held at 20 degrees C. **The insulated vessel cools more slowly.** The chart and measurements below describe the same cooling model.

Experimental setup

Each vessel contains the same water volume. A probe records temperature every five minutes. Keep the lid position and room conditions unchanged.

1. Record the initial temperature before starting the timer.
2. Keep both probes at the same depth.
3. Read the measurements without moving either vessel.

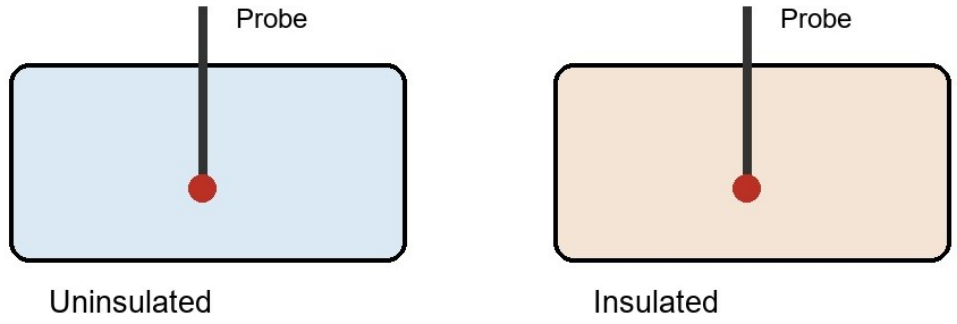


Figure 1 Probe positions in the two vessels

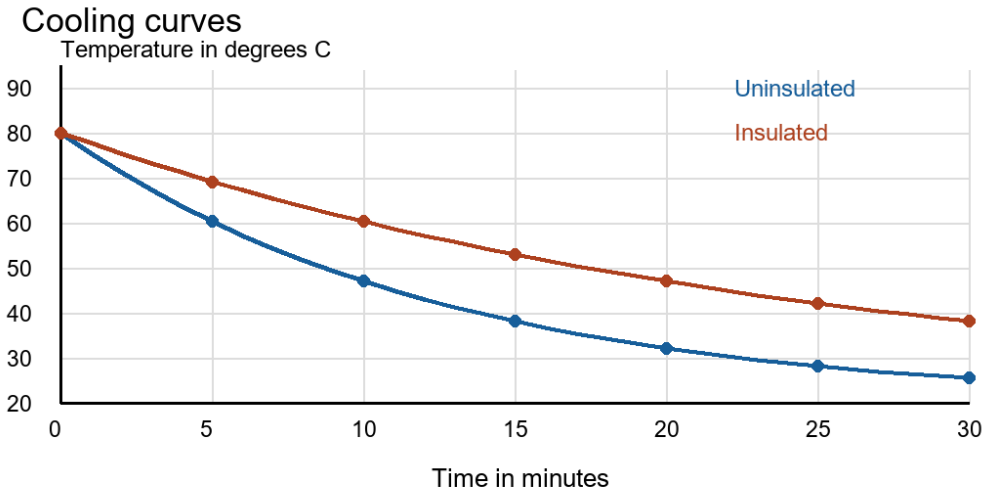


Figure 2 Cooling curves over thirty minutes

Model and measurements

Newton cooling model

$$T(t) = T_a + (T_0 - T_a)e^{-kt}$$

Half cooling time

$$t_{1/2} = \frac{\ln 2}{k}$$

The rate constants are 0.08 and 0.04 per minute. The half cooling times are approximately 8.66 and 17.33 minutes respectively.

Recorded temperatures

Water temperature in degrees C		
Time in minutes	**Uninsulated**	**Insulated**
0	80.0	80.0
5	60.2	69.1
10	47.0	60.2
15	38.1	52.9
20	32.1	47.0
25	28.1	42.1
30	25.4	38.1

Table 1 Measurements at five minute intervals

Interpretation

After thirty minutes, the insulated vessel remains approximately 12.6 degrees C warmer.[1]

Review decision: *repeat the experiment with a second probe*

END OF REPORT

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